vallourec

2669 Martin Luther King Jr Blvd
Youngstown,OH 44510

VALLOUREC STAR LP

Seamless Tubular Products

Certified Test Report
01/02/2026
Revision: 0

Order: Y-221130650-001 **Desc:** Seamless Line Pipe- Normalized Rolled **Customer:** Kelly Pipe Company **SLN / (Heat):** PN0251

Grade: API 5L X52N/LT Norm PSL2 46th Edition May 01, 2019 **Customer Order:** 10P46500 **Split:** 01

OD(in): 6.625 **Wall(in):** 0.280 **Lbs/Ft:** 18.99 **Length:** Double Random

718543 PN0251

End Finish: Plain End Beveled

Comments: Melted and Manufactured in Youngstown, Ohio USA

Vallourec Star's QA program meets the requirements of Directive 2014/68/EU and DIN EN 10204-2004 Type 3.1. Meets the requirements of ASTM A106B-19a, A333 Gr.1 and 6-24 and A53B-24; ASME SA A333 Gr.1 and 6: 2025, SA106B:2025, and SA53B: 2025. Satisfies NACE MR0175-2021/15156-1:2020 and ANSI/NACE MR0103/ISO17945:2015(R2023). Material has been produced using silicon killed carbon steel with aluminum fine grain practice. No repair welding has been conducted. Material is free from Mercury. API MONOGRAMMED

Mechanical Properties

		Tensile Size Specimen Cross Section			Strength (ksi)			Elongation		
Orientation	**Type**	**Width (in)**	**Thick (in)**	**Area (sqin)**	**Yield**	**Tensile**	**Y/T Ratio**	**Gage Length**	**% Elong**	**% RA**
Longitudinal	Strip	1.000	0.290	0.2900	60.3	81.6	0.74	2	35.2	67.8
Longitudinal	Strip	1.000	0.300	0.3000	59.8	80.7	0.74	2	37.3	68.6
Longitudinal	Strip	1.000	0.290	0.2900	58.0	79.6	0.73	2	37.8	66.4

Required Min/Max Yield Strength (ksi): 52.2/76.9 **Required Min/Max Tensile Strength (ksi):** 70/110.2 **Calculated Max Internal Yield (psi):** 4292

Required Min Elongation(%): 27

Supplemental Requirements

Flattening Test	Passed
Hydrostatic Pressure	3310 psi
Hydrostatic Hold Time	5 seconds

Chemical Analysis **Chem Spec:** 69

WT%	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Sn	Nb	V	Al	Ca	B	Ti	N	C.E.
Heat	0.17	1.23	0.010	0.005	0.30	0.10	0.04	0.11	0.01	0.007	0.000	0.054	0.024	0.0022	0.0005	0.004	0.0065	0.42
Product 3	0.17	1.21	0.010	0.003	0.28	0.09	0.04	0.10	0.01	0.007	0.002	0.056	0.023	0.0018	0.0001	0.003	0.0068	0.41
Product 4	0.17	1.22	0.010	0.003	0.29	0.10	0.04	0.10	0.01	0.006	0.002	0.056	0.024	0.0019	0.0001	0.004	0.0060	0.42

Charpy Impact Testing

	Notch	Direction	Size	Temp °F	Ft Lbf 1	Ft Lbf 2	Ft Lbf 3	Ft Lbf Avg	Shear 1	Shear 2	Shear 3	Shear Avg
Actual	V	Longitudinal	1/2	-70	53	59	64	59	40	40	40	40

This material has been produced and tested in accordance with the requirements of applicable specifications unless otherwise listed below. We hereby certify that the above test results are representative of those contained in the records of the company. Any modification to this certification as provided by Vallourec Star without the expressed written consent of Vallourec Star negates the validity of this test report. Vallourec Star is not responsible for the inability of this material to meet specific applications.

THIS CERTIFICATE IS NOTARIZED ONLY WHEN REQUESTED.

Signed: Jeff Schultz

Name: Jeff Schultz, Lab Supervisor

Date Approved: 01/05/2026

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Inspection Company: L Bay

Customer: Kelly Pipe Company
Customer Order: 10P46500

SLN / (Heat): PN0251
Split: 01

Non-Destructive Evaluation Criteria

UT CHANNELS
ID
10% Transverse
10% Longitudinal
OD
5% Transverse
5% Longitudinal

EMI CHANNELS
ID
10% Longitudinal
OD
5% Longitudinal

FULL LENGTH WALL THICKNESS CHECK
87.5% Wall Reduction
Required Volumetric Coverage: 100%

ADDITIONAL REQUIREMENTS
Notch Length < / = 1.00"
Notch Width .020" Max
Untested Ends Cropped
OD
Visual Insp. Full Length - White Light
ID
Visual Insp. Pipe Ends - White LIght

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Mill Test Certification

Hardness Test Results

SLN PN0251 **Lot** As Rolled **OD(in)** 6.625 **Wall(in)** 0.280

Hardness 3 Point

Test Date	Joint	Sample	Scale	Position	OD	MID	ID	AVG
12/14/2025 12:12:00 PM		AS ROLLED	HRBW	Lead	83.8	83.3	83.7	83.6
12/14/2025 02:19:50 PM		AS ROLLED	HRBW	Lead	84.1	84.2	84.2	84.2
12/14/2025 12:13:15 PM		AS ROLLED	HRBW	Lead	83.2	83.2	83.6	83.3
12/14/2025 12:12:47 PM		AS ROLLED	HRBW	Lead	82.3	82.8	82.8	82.6